Ex 2: Data Discovery and Preparation

Question

(An Autonomous Institution, Affiliated to Anna University, Chennai)
Department of Artificial Intelligence
V Semester B. Tech AIDS
AD23532 – Principles of Data Science

Ex2: Data Discovery and Preparation

Learning Objective:
The objective of this experiment is to perform data discovery and exploratory data analysis on a real-world dataset.

Question:
A telecommunications company has observed an increase in customer churn, leading to revenue loss. The management wants to understand customer behavior and identify patterns before developing a machine learning model to predict churn.

As a Data Analyst, your responsibility is to discover meaningful insights from the dataset, identify data quality issues, perform exploratory data analysis, detect outliers, and prepare the dataset for predictive analytics.

Tasks to Perform:

1. Reuse Ex:1 Steps 1- 6, 10.
2. Analyze the distribution of: Customer Churn, Gender, Contract Type, Payment Method.
3. Visualize the distribution of: MonthlyCharges, TotalCharges Tenure using Histograms and Boxplots.
4. Detect outliers in numerical features using the Interquartile Range (IQR) method.
5. Create suitable visualizations (Count Plot, Box Plot, Histogram, or Scatter Plot) to study the relationship between: Contract Type and Churn, Internet Service and Churn, Monthly Charges and Churn, Tenure and Churn.
6. Do the below analytical questions:
   - Which contract type has the highest customer churn?
   - Which payment method is most frequently used by customers who churn?
   - Do customers with shorter tenure tend to churn more often?
   - Which customer group has the highest monthly charges?
   - Which Internet Service category experiences the highest churn?
   - Is there any relationship between monthly charges and churn?
   - Which features appear to be the strongest indicators of customer churn?
7. Split the cleaned dataset into Training (80%) and Testing (20%) datasets.
8. Save the final processed dataset as a new CSV file named CustomerChurn_Preprocessed.csv.

Aim

To perform data discovery and exploratory data analysis on the telecom churn dataset, detect outliers using the IQR method, identify the strongest indicators of churn and prepare the train/test datasets.

Procedure

1. Reuse the Ex1 steps 1–6 and 10 (import, load, explore, clean, encode, scale, correlation heatmap).
2. Analyse the distribution of Churn, Gender, Contract and PaymentMethod.
3. Plot histograms and box plots of MonthlyCharges, TotalCharges and Tenure.
4. Detect outliers in each numerical feature using the IQR rule (Q1 − 1.5·IQR, Q3 + 1.5·IQR).
5. Visualise Contract vs Churn, InternetService vs Churn, MonthlyCharges vs Churn and Tenure vs Churn.
6. Answer the analytical questions from the plots and group-by tables.
7. Split the data 80:20 into training and testing sets and save the processed data as CustomerChurn_Preprocessed.csv.

Code

```
#Mohammed Fayyad(241801165)
# (Run the Ex1 cleaning, encoding and scaling code first; 'raw' = cleaned, un-
encoded data)
from sklearn.model_selection import train_test_split

# Distributions
for c in ["Churn", "gender", "Contract", "PaymentMethod"]:
    print(raw[c].value_counts(normalize=True) * 100)
    sns.countplot(x=c, data=raw); plt.xticks(rotation=30); plt.show()

# Histograms and box plots
for c in ["MonthlyCharges", "TotalCharges", "tenure"]:
    fig, ax = plt.subplots(1, 2, figsize=(10, 4))
    sns.histplot(raw[c], kde=True, ax=ax[0]); sns.boxplot(x=raw[c], ax=ax[1])
    plt.show()

# IQR outliers
for c in ["MonthlyCharges", "TotalCharges", "tenure"]:
    q1, q3 = raw[c].quantile([0.25, 0.75]); iqr = q3 - q1
    out = raw[(raw[c] < q1 - 1.5*iqr) | (raw[c] > q3 + 1.5*iqr)]
    print(c, "outliers:", len(out))

# Relationships with churn
sns.countplot(x="Contract", hue="Churn", data=raw); plt.show()
sns.countplot(x="InternetService", hue="Churn", data=raw); plt.show()
```

```
sns.boxplot(x="Churn", y="MonthlyCharges", data=raw); plt.show()
sns.boxplot(x="Churn", y="tenure", data=raw); plt.show()

# Analytical questions
print(pd.crosstab(raw["Contract"], raw["Churn"], normalize="index"))
print(raw[raw.Churn == "Yes"]["PaymentMethod"].value_counts())
print(raw.groupby("Churn")[["tenure", "MonthlyCharges"]].mean())
print(raw.groupby("InternetService")["MonthlyCharges"].mean())
print(pd.crosstab(raw["InternetService"], raw["Churn"], normalize="index"))
print(df.select_dtypes(include=np.number).corr()["Churn"].sort_values())

# Split and save
X = df.drop(columns=["Churn", "customerID"], errors="ignore")
y = df["Churn"]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42, stratify=y)
print(X_train.shape, X_test.shape)
df.to_csv("CustomerChurn_Preprocessed.csv", index=False)
```

Result

Exploratory analysis, outlier detection and relationship plots were completed, the analytical questions were answered from the outputs, the data was split 80:20, and the processed dataset was saved as CustomerChurn_Preprocessed.csv.

